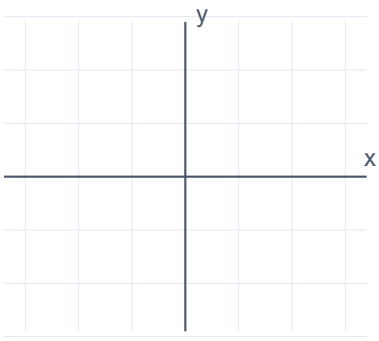
Reference angles and exact values

Core practice · Page 1 of 2 · Name: _____

Sketch each angle. Give exact values and show your reasoning. These are fresh practice examples; quiz dates and official coverage are in Canvas.

1. For θ = 240°, sketch the terminal side and complete the chain.

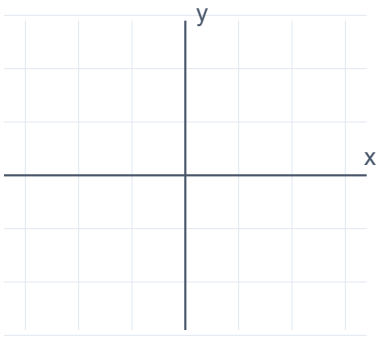


Quadrant: _____

Reference angle α = _____

sin α = _____; sin θ = _____

2. For θ = −7π/4, sketch the terminal side and complete the chain in radians.



Quadrant: _____

Reference angle α = _____

cos α = _____; cos θ = _____

3. Identify the quadrant and explain how both sign conditions hold.

(a) sec θ > 0 and tan θ < 0: _____

(b) sin θ < 0 and tan θ > 0: _____

A signed point and exact equations

Core practice · Page 2 of 2 · Work first, then check the separate answer key.

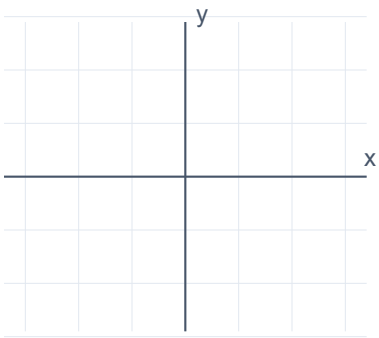
4. The terminal side of θ passes through (−4, −7). Find r and all six trigonometric functions. Keep radicals exact.

r = √(_____ + _____) = _____

sin θ	cos θ	tan θ
csc θ	sec θ	cot θ

5. Find all solutions with 0° ≤ θ < 360°. Give exact angle measures.

(a) cos θ = −1/2



Reference angle: _____

Both solution angles: _____

(b) sin θ = √2/2

Reference angle: _____ Both solution angles: _____

Optional calculator and domain checks

Extension practice · Complete after the two core pages or use for extra review.

6. A second terminal point is (−8, −14). Would the six function values change? Explain without starting over.

7. Find all solutions of sin θ = −0.620 with 0° ≤ θ < 360°. Label the calculator's principal output separately and round only the final answers to two decimals.

8. Find all solutions of tan θ = −1.400 with 0° ≤ θ < 360°. Round final answers to two decimals.

9. Does sin θ = 1.120 have any real solution? Explain before using a calculator.

10. Find sin 270°, cos 0°, tan 270°, and csc 180°. Identify any zero denominator.

Answer key

Core pages first; optional extension answers follow. Equivalent radical forms are valid.

1. 240° is in QIII. α = 60°; sin α = √3/2; sin θ = −√3/2.

2. −7π/4 + 2π = π/4, in QI. α = π/4; cos α = √2/2; cos θ = √2/2.

3. (a) QIV: secant and cosine are positive, while tangent is negative. (b) QIII: sine is negative and tangent is positive.

4. r = √(16 + 49) = √65. sin θ = −7/√65; cos θ = −4/√65; tan θ = 7/4; csc θ = −√65/7; sec θ = −√65/4; cot θ = 4/7. Rationalized equivalents are acceptable.

5. (a) Reference angle 60°; cosine is negative in QII and QIII: 120°, 240°. (b) Reference angle 45°; sine is positive in QI and QII: 45°, 135°.

6. No. The new point is twice as far from the origin on the same ray. x, y, and r all double, so every ratio stays the same.

7. Principal output ≈ −38.32°, outside the interval. Sine is negative in QIII and QIV. The interval solutions are 218.32° and 321.68°.

8. The reference angle is arctan(1.400). Tangent is negative in QII and QIV. The interval solutions are 125.54° and 305.54°.

9. No real solution. Every real sine value lies in [−1, 1].

10. sin 270° = −1; cos 0° = 1. tan 270° is undefined because x = 0; csc 180° is undefined because y = 0.